

HP-200 Service Manual

Hydraulic Press HP-200X Heavy Duty — Rev 4.2 (2024)

This document is a demonstration OEM-style manual generated for the SARVA-SENSE RAG troubleshooting assistant prototype.

1-5.1 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.2 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.3 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.4 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.5 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.6 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.7 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.8 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.9 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.10 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.11 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.12 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.13 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.14 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.15 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.16 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.17 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.18 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.19 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.20 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.21 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.22 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.23 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.24 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.25 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.26 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.27 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.28 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.29 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.30 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.31 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.32 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.33 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.34 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.35 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.36 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.37 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.38 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.39 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.40 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.41 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.42 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.43 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.44 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.45 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.46 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.47 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.48 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.49 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.50 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.51 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.52 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.53 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.54 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.55 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.56 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.57 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.58 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.59 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.60 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.61 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.62 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.63 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.64 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.65 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.66 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.67 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.68 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.69 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.70 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.71 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.72 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.73 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.74 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.75 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.76 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.77 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.78 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.79 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.80 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.81 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.82 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.83 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.84 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.85 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.86 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.87 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.88 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.89 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.90 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.91 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.92 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.93 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.94 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.95 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.96 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.97 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.98 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.99 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.100 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.101 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.102 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.103 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.104 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.105 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.106 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.107 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.108 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.109 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.110 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.111 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.112 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.113 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.114 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.115 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.116 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.117 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.118 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.119 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.120 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.121 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.122 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.123 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.124 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.125 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.126 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.127 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.128 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.129 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.130 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.131 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.132 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.133 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.134 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.135 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.136 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.137 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.138 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.139 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.140 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.141 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.142 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.143 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.144 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.145 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.146 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.147 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.148 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.149 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.150 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.151 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.152 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.153 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.154 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.155 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.156 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.157 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.158 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.159 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.160 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.161 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.162 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.163 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.164 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.165 General Reference Note

Refer to OEM standard reference material for this subsystem.

1-5.166 General Reference Note

Refer to OEM standard reference material for this subsystem.

SECTION 6.1 - THERMAL MANAGEMENT & HEAT EXCHANGERS

Hydraulic Thermal Limits: normal operating zone is 38°C to 55°C measured at reservoir sight glass LG-01. A warning alarm triggers at 60°C. Continuous fluid temperatures exceeding 60°C rapidly degrade seals and accelerate oil oxidation. A critical automatic machine shut-off trips at 68°C.

Probable causes of hydraulic over-temperature: (1) oil-to-air heat exchanger HE-01 fins clogged with airborne particulate and coolant mist, (2) hydraulic reservoir oil level dropped below the minimum indicator, reducing thermal dissipation mass, (3) system pressure relief valve RV-01 stuck partially unseated, generating continuous parasitic bypass friction and heat, (4) circulation pump cooling fan motor relay failure.

Corrective Action Procedure: Step 1 - stop the press stroke cycle immediately and leave circulation fans running for 10 minutes if operational (THERMAL BURN HAZARD: hydraulic fluid and manifold surfaces may exceed 75°C, wear heat-resistant nitrile gloves and eye protection). Step 2 - inspect oil sight glass and level gauge LG-01; fluid must sit between the upper and lower index marks at 40°C. Step 3 - blow compressed air (below 2 bar) from inside outward through heat exchanger HE-01 fins to clear debris. Step 4 - use an infrared thermal gun on relief valve RV-01 body; if its temperature exceeds reservoir temperature by more than 15°C, the valve is leaking internally to tank and must be rebuilt or replaced.

6-7.1 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.2 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.3 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.4 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.5 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.6 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.7 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.8 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.9 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.10 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.11 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.12 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.13 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.14 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.15 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.16 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.17 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.18 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.19 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.20 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.21 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.22 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.23 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.24 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.25 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.26 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.27 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.28 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.29 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.30 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.31 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.32 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.33 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.34 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.35 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.36 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.37 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.38 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.39 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.40 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.41 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.42 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.43 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.44 General Reference Note

Refer to OEM standard reference material for this subsystem.

6-7.45 General Reference Note

Refer to OEM standard reference material for this subsystem.

SECTION 8.3 - HYDRAULIC DIAGNOSTIC & ALARM MATRIX

ALARM CODE E101: Hydraulic Pressure Sensor Fault, sub-circuit Primary Manifold Block A. Trigger condition: PLC analog input 04 loop current under 3.6 mA or over 21.5 mA for more than 450 ms. Severity: Critical, automatic protective stop and emergency ramp-down.

Probable root causes of E101: (1) pressure transducer PX-102 M12 connector unplugged, bent pins, corroded or damaged by fluid ingress, (2) signal wiring 31/32 open circuit between junction box JB-2 and PLC analog module 04, (3) transducer diaphragm mechanical blowout, (4) hydraulic system pressure genuinely dropped below the 140 bar minimum threshold.

Corrective Action Procedure for E101: Step 1 - engage Lock-Out/Tag-Out (LOTO) on main isolator SW-1 and depressurize the accumulator via manual bleed valve BV-01 (HIGH PRESSURE HAZARD: fluid injection injury can be fatal, never loosen manifold fittings while the accumulator is charged). Step 2 - remove the M12 connector on manifold block A and inspect pins for bending, moisture, or thermal degradation; clean with electrical contact cleaner. Step 3 - connect a digital multimeter in series across terminals 31 (+) and 32 (-); expected nominal loop current in standby is 4.02 mA +/- 0.05 mA. Step 4 - cross-check analog dial gauge G-101 against the digital display; if the manual gauge reads normal pressure above 160 bar, replace transducer PX-102 (part no. HYD-TR-8821).

ALARM CODE E102: Hydraulic Return Line Filter Clogged. Trigger: differential pressure switch DPS-2 held closed for over 30 seconds. Corrective action: replace return filter element RF-220 and reset the differential switch.

ALARM CODE E105: Accumulator Pre-charge Low. Trigger: nitrogen pre-charge pressure sensor reads below 55 bar during idle cycle. Corrective action: schedule accumulator nitrogen recharge per Section 8.5, do not operate press until pre-charge is restored.

8.1 General Reference Note

Refer to OEM standard reference material for this subsystem.

8.2 General Reference Note

Refer to OEM standard reference material for this subsystem.

8.3 General Reference Note

Refer to OEM standard reference material for this subsystem.

8.4 General Reference Note

Refer to OEM standard reference material for this subsystem.

8.5 General Reference Note

Refer to OEM standard reference material for this subsystem.